

Paper Review Synopsis

Date: 2026-09-07 **Purpose:** Single-document review of the papers that carry the argument. Two tables: Table A covers the 18 primary comparators with deep prose per paper (what it does · our comment · why we selected it); Table B covers the 86 extended-corpus papers at index depth (category · role in the review). Each row also links to the paper's full extraction notes. **Audience:** Supervisor / reviewer wanting to see the whole review in one artefact rather than distributed across chapter prose, individual notes files, and the master coding table. **Status:** Complete. Table A drafted 2026-09-07 (18 rows, deep); Table B added later the same day (86 rows, grouped by nine thematic categories, shallow-index depth).

Table A — Primary comparators (18)

The 18 systems from Chapter 2 Table 2.1, alphabetical by first author. Each row carries three things: a mechanism description in *our* framing (not the paper's own), an interpretive comment naming what is right / what is missing relative to our architecture / why the missing part matters, and the selection rationale.

Paper	What the paper does	Our comment	Why we selected it
Abella et al. (2025) — SAFEXPLAIN [notes]	EU-project reference architecture that implements a supervision function bounding AI operations within a predefined safety envelope, with automatic switching to a non-AI fallback subsystem when the envelope is breached.	Envelope-based binary switching in an automotive / space context. The envelope is defined at design time, not derived from classified environmental state, so the switching decision cannot vary in kind, only in when it fires. The fallback is a non-AI system, which means our intermediate CAUTION mode (AI participating under restricted scope) has no analogue in the SAFEXPLAIN architecture.	Comparator for the "AI operates within predefined envelope or is replaced" pattern; one of the most institutionally credible references for envelope-based governance.
Bajcsy & Fisac (2024) [notes]	Defines the Human-AI Safety Filter as a control-theoretic mechanism that monitors AI outputs and overrides them at the last possible moment to prevent trajectory-level safety violations.	Sophisticated Level-1 gate — the ceiling of what participation control can achieve without touching advisory scope. The formal safety proof holds only for binary intervention; it does not extend to graduated governance because the codomain is {intervene, permit}, not {SAFE, CAUTION, UNSAFE}.	Establishes the sophistication ceiling of Level-1 participation control against which our L2 contribution is contrasted.
Banerjee et al. (2025) —	Introduces an augmented grammar	Structurally the closest formal analogue in the corpus to our	Nearest formal precursor to the

Paper	What the paper does	Our comment	Why we selected it
CRANE [notes]	$G_a = R_M \cdot G$ that alternates between unconstrained reasoning generation and constrained output generation, and formally proves (Proposition 3.1) that binary output constraint reduces LLM expressivity to the TC^0 complexity class.	A_{AI} containment relationship: the constrained output set is a strict subset of the unconstrained set, exactly as $A_{AI}(CAUTION) \subset A_{AI}(SAFE)$. The alternation is triggered by syntactic delimiter tokens rather than classified environmental state, so the conditioning variable differs. Proposition 3.1 is separately valuable as a computational-theoretic argument for why binary output constraint is suboptimal.	containment structure of A_{AI} , and the only paper in the corpus offering a complexity-theoretic proof that binary output constraint degrades AI capability.
Baxi (2026) — CGAE [notes]	Formalises a K-tier permission architecture for AI economic agents, with nested permission sets $E(T_k) \subset E(T_{k+1})$ across discrete tiers, conditioning each agent's tier on its worst verified robustness dimension.	Structurally the nearest formal comparator in the corpus. The nested-permission containment is a direct analogue of $A_{AI}(SAFE) \supset A_{AI}(CAUTION) \supset A_{AI}(UNSAFE)$. The critical difference is the conditioning variable: Baxi conditions on offline-verified agent robustness, we condition on classified environmental state. Baxi answers "how much can this AI be trusted to do?"; we answer "what is the AI permitted to do given current environmental conditions?" These are orthogonal questions.	Would be the strongest counter-example to our novelty claim without the conditioning-variable distinction; including it is essential to preclude the "isn't this just CGAE?" objection.
Chen, Kang & Li (2025) — SHIELDAGENT [notes]	Extracts LTL rules from policy documents and uses formal verification to evaluate each planned agent action against the derived rule circuits, producing a binary safe / unsafe label per action.	Policy-driven per-action validation, sophisticated in its formal-methods stack. The governance operates at the individual-action level with no restriction of the agent's action type space: the shield fires per action, and the recommendation space is not restricted before reasoning.	State-of-the-art in LTL-based policy shielding; illustrates the pattern of per-action validation that does not condition on environmental state.
Corsi et al. (2024) —	Refines the shielding paradigm by using	Shield refinement improves efficiency but retains binary	Demonstrates that even sophisticated

Paper	What the paper does	Our comment	Why we selected it
Verification-Guided Shielding [notes]	formal DNN verification to partition the input space into provably safe and potentially unsafe regions, activating the shield only in the verified-unsafe regions and reducing overhead by 25–71%.	character: the shield either intervenes or permits, with no intermediate response. The verified-safe / potentially-unsafe partition is a preprocessing step, not a runtime governance mode.	shield refinements do not transcend binary governance; anticipates the "but modern shields are more nuanced" objection.
Dalrymple et al. (2024) — Guaranteed Safe AI [notes]	Proposes the Guaranteed Safe AI framework built on three components: a world model, a safety specification, and a verifier that produces auditable proof certificates before AI output is deployed.	GS AI is the theoretical umbrella our architecture instantiates. The framework itself is binary at the verification level: proof certificates either verify or fail. Our architecture is a state-conditioned instantiation of GS principles that fills a gap the framework leaves open — how to govern under conditions that are elevated but not prohibitive.	The closest theoretical precedent; positioning our architecture as a domain-specific instantiation of GS principles is a core framing decision.
Feng, McDonald & Zhang (2025) — Levels of Autonomy [notes]	Proposes a five-level autonomy framework for AI agents with two governance dimensions (agency and autonomy) and introduces "autonomy certificates" as formal credentials for agent capabilities.	The most conceptually granular authority-allocation framework in the corpus. Both dimensions are configured at design time and do not vary with operational conditions — a Level-3 agent stays Level-3 regardless of whether the environment is safe or dangerous. Their two-dimension decomposition makes it easy to state precisely what our contribution adds: advisory-scope restriction is a missing third governance lever.	Most granular authority-allocation framework identified; its explicit two-dimension decomposition makes the missing-third-lever argument concrete rather than abstract.
Flehmgig et al. (2024) [notes]	Implements a traffic-light degradation index for AI in safety-critical systems, classifying AI performance into three levels (green /	The closest structural precedent — three-level classification triggered by monitoring, with an intermediate level. The intermediate level governs supervisory intensity (how closely the human watches), not	The structurally closest precedent in the corpus; makes the "why is your CAUTION mode different from Flehmig's orange?"

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	orange / red) using non-AI statistical methods; at orange, the supervisory component conducts thorough checks; at red, the system switches to a non-AI backup.	advisory scope (what the AI may recommend). The AI's output is identical at green and orange. Central to our F-15 empirical finding: a Flehmig-style baseline is output-equivalent to a plain binary gate (0.00% divergence over 5 years of hourly data), which converts the CAUTION-mode gap from a conceptual claim into an empirical measurement.	question inescapable, and gives our empirical F-15 result its comparison target.
Gyllenhammar et al. (2025) [notes]	Reviews safety-evidence methods for automated driving, identifying a three-tier hierarchy of Operational Design Domain (ODD), Restricted Operational Domain (ROD), and Minimal Risk Condition (MRC).	Three-level degradation hierarchy — closest to graduated governance in the automated-driving literature — but the intermediate ROD restricts <i>where</i> the vehicle operates (geographic domain confinement), not <i>what</i> recommendations the system produces within its operational scope. Operational-domain confinement is a different governance object from advisory-scope restriction.	Comparator for domain-confinement adaptation; shows that "three levels" in adjacent literature typically restricts operational scope rather than advisory type space.
Hamel-De le Court, Belardinelli & Goodall (2025) [notes]	Extends the shielding paradigm to stochastic environments, constructing a probabilistic shield with two formal guarantees — Theorem 1 (shielded policy satisfies safety constraints with probability $\geq 1 - \delta$) and Theorem 2 (convergence of the underlying learning algorithm is preserved).	The most formally rigorous shield extension to stochastic settings, yet the intervention mechanism remains binary: the shield either intervenes when estimated risk exceeds a threshold or permits the action. Probabilistic formulation of the safety guarantee does not change the binary character of the governance decision.	Rules out the "but modern shields handle stochasticity, which gives graduated behaviour" argument; the guarantee is probabilistic but the decision is not.

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Könighofer et al. (2025) [notes]	Formalises shields as runtime mechanisms synthesised from LTL safety specifications and environment models that intercept unsafe actions before they reach the environment.	The canonical reference for the shield formalism. Every shield-derived governance mechanism in the corpus builds on this. The mechanism is binary by construction: the shield either allows the action or replaces it with a safe alternative; no intermediate mode is definable within the framework.	The reference implementation of Level-1 binary governance; any comparison to shielding starts here.
Kwon et al. (2025) [notes]	Adaptive shielding framework that incorporates environment-centric features $E \in \mathbb{R}^{n^2}$ into a SafetyScore driving runtime shield adjustment via an Adaptive Constraint Parameter (ACP) threshold Γ_t ; Theorem 1 provides a formal safety guarantee that the shielded policy satisfies constraints throughout the adaptation.	The closest near-miss in the corpus: environment-centric features as governance input, runtime adaptation, and a formal safety proof — all three properties we claim. Two critical differences remain. First, the SafetyScore is binary (each state-action pair is classified safe or unsafe, with no intermediate). Second, the adaptation is quantitative (ACP threshold Γ_t shifts along a continuous scale) rather than qualitative (no discrete governance modes analogous to SAFE, CAUTION, UNSAFE). Kwon adapts <i>how strictly</i> the shield intervenes; we adapt <i>what the AI is permitted to recommend</i> .	Would be the strongest technical counter-example ("this paper already uses environment as governance input") without our binary-vs-graduated and quantitative-vs-qualitative distinctions; retained to preclude that objection.
Odriozola-Olalde et al. (2023) [notes]	Systematic review of shielded reinforcement learning classifying methods into three Safety Levels according to how the shield interacts with the policy.	The three Safety Levels sound graduated but all three are binary per-action gates — the classification is about implementation stack (shield-then-policy vs shield-and-policy vs policy-only-with-safety), not about the governance decision itself. Confirms that "graduated" in shielded RL means implementation-tier gradation, not advisory-scope gradation.	Anchor-adjacent SLR that pre-empts the reading "surely shielded RL literature already has three-level governance" — it does not, in the sense we mean.

Paper	What the paper does	Our comment	Why we selected it
Perez-Cerrolaza et al. (2024) [notes]	Survey of AI safety techniques across industrial and transportation domains (294 references), classifying AI safety compliance into three classes and cataloguing safety bags, monitors, and envelopes as dominant runtime mechanisms.	Anchor SLR. Establishes binary participation governance across the industrial and transportation literature; the three compliance classes describe how the AI is certified, not what it may recommend. The scale of the review (294 references) is what makes it evidentially strong — any counter-example would have to hide in a corpus this large without leaving a trace.	One of the three anchor systematic reviews that make the absence claim credible; without it, our claim would be a survey of primary comparators rather than a structural finding across the literature.
Ramos et al. (2024) [notes]	Systematic review of 91 studies on collaborative intelligence in safety-critical industries, identifying two primary interaction modes (machine-assists-human and human-assists-machine) across six industries.	Anchor SLR. Finds binary governance is universal across 91 papers spanning six industries — machine-assists-human is the dominant pattern. The size and industry breadth are what make the finding structural rather than incidental to any single sector.	Anchor SLR confirming binary governance from the collaborative-intelligence literature; refutes the "surely someone has done this in the last 91 papers" objection.
Shamsujjoha et al. (2025) — Swiss Cheese Model [notes]	Multi-layered guardrail taxonomy for foundation-model agents, cataloguing 13 distinct guardrail actions across 14 pipeline target categories.	The definitive Level-2 (output-restriction) taxonomy — the most comprehensive catalogue of what modern guardrails can do. None of the 13 actions conditions on classified environmental state; "context-dependent rules" are one of four categories but refer to deployment context (user location, regulatory jurisdiction), not runtime environmental safety state. Establishes that even the most complete guardrail vocabulary lacks the state-conditioning that defines our contribution.	The reference source for the strongest form of Level-2 governance in the corpus; if any single paper could refute our L2-plus-state-conditioning claim from the guardrail side, this would be it.

Paper	What the paper does	Our comment	Why we selected it
Wang et al. (2026) — AgentSpec [notes]	Domain-specific language for runtime enforcement over LLM agents, allowing developers to define structured rules composed of triggers, predicates, and enforcement actions; prevents over 90% of unsafe code executions in the reported evaluation.	State-of-the-art in per-action runtime enforcement. The rule set is fixed at deployment; predicates evaluate the AI's proposed action against those rules, not against a classified environmental state. Illustrates that even sophisticated per-action enforcement remains invariant to environmental conditions.	Represents the most sophisticated per-action enforcement pattern in the corpus; comparator for the claim that even DSL-configured enforcement does not condition on runtime state.

Table B — Extended-corpus papers (86)

The 86 substantively-extracted corpus papers that are not among the 18 primary comparators in Table A. Rows are grouped by thematic category (nine groups) so the table skims by theme; within each group, rows are alphabetical by first author. Each row's *Role in the review* summarises the paper's argumentative function; open the linked notes for the full extraction and per-paper analysis.

Coverage: 76 of the 86 rows carry a finding pulled from the paper's own extraction notes (Bottom Line, Positioning, Key Findings, or Core Contribution sections). The remaining 10 rows carry a generic pointer to the notes — those are papers whose extraction notes do not surface a finding in a finding-bearing section outside their table content. Follow the notes link to see the full analysis for any such row.

AI governance frameworks (26)

Governance frameworks that establish the general shape of the problem or provide contextual comparators. Included to anchor the review in institutional and regulatory literature adjacent to the technical comparators in Table A.

Paper	Role in the review
Attard-Frost & Lyons (2025) [notes]	This paper is a high-quality, peer-reviewed source that you can use to establish what institutional AI governance looks like at maximum empirical resolution — and to show that even at that resolution, the concept of state-conditioned runtime advisory scope restriction does not appear.
Batool et al. (2025) [AI governance SLR, external evidence] [notes]	Positioning paragraph: Batool, Zowghi, and Bano (2025), systematically reviewing 28 AI governance studies, find coverage to be fragmentary — only three studies address who governs, what is governed, when, and how — with solutions concentrated at the organizational level and oriented toward compliance and risk management rather than enforceable principle.
Bello y Villarino et al. (2025) [notes]	Problem: Emerging AI regulatory frameworks (EU AI Act, NSW AIAF) restrict their scope to systems classified as "AI," but this may leave many

Paper	Role in the review
	high-risk automated decision-making (ADM) systems outside regulatory oversight — particularly rule-based systems operating in sensitive public sector domains.
Bengio et al. (2026) [Intl AI Safety Report] [notes]	Main problem addressed: The "evidence dilemma" facing policymakers — general-purpose AI capabilities are evolving rapidly, but evidence on risks is slow to emerge and difficult to assess.
Bengio, Hinton et al. (2024) [Science] [notes]	Problem: AI is progressing rapidly toward generalist, autonomous systems. Risks include large-scale social harms, malicious uses, and irreversible loss of human control.
Bloomfield & Rushby (2025) [notes]	Bloomfield & Rushby (2025) represents the most comprehensive treatment of AI assurance from the dependability perspective, providing a systematic taxonomy of guarded architectures for safety-critical AI systems.
Castagnone & Nitti (2026) [notes]	Problem: LLMs are fundamentally unsuitable for direct application in structural engineering due to their probabilistic nature, hallucination risk, and non-deterministic outputs.
Chen et al. (2025) [LLM Safety Survey] [notes]	Problem addressed: The AI safety research landscape lacks a unified, structured framework that integrates the technical (Trustworthy AI), ethical (Responsible AI), and ecosystem-level (Safe AI) dimensions of AI safety into a coherent architectural view.
Engin & Hand (2025) [dimensional governance, external evidence] [notes]	Positioning paragraph: Engin and Hand (2025) argue from the governance-theory literature that static categorical frameworks — fixed risk tiers, autonomy levels, oversight models — are increasingly insufficient for dynamic AI systems, and propose dimensional governance: categories repositioned as explicit thresholds over continuously monitored dimensions, adjusted as conditions evolve, with system ...
Kamath et al. (2025) [POD-Attention, external evidence] [notes]	Positioning paragraph: Kamath et al. (2025) provide peer-reviewed systems-level evidence that LLM inference follows a structurally fixed two-phase execution: a compute-bound prefill phase that processes the entire prompt, followed by a memory-bandwidth-bound decode phase that generates output tokens one at a time, autoregressively.
Kang (2026) [notes]	Problem: Agentic AI coding systems in regulated industries (banking, insurance, healthcare) face a "Productivity-Reliability Paradox": uniform human review of everything collapses velocity; reviewing nothing violates regulatory human-oversight expectations.
Kolt et al. (2025) [complex systems AI governance, external evidence] [notes]	Positioning paragraph: Kolt, Shur-Ofry, and Cohen (2025) argue from complex systems science that AI governance premised on linear cause-and-effect is structurally inadequate: because AI systems exhibit nonlinear growth, emergence, and cascading effects, governance must intervene

Paper	Role in the review
	early and at calibrated risk thresholds, acting on incomplete information — since by the time full evidence of danger is...
Koohestani (2025) [AgentGuard, runtime verification, workshop PoC] [notes]	Positioning paragraph: Koohestani (2025) represents the current frontier of the runtime verification line: AgentGuard replaces static pre-deployment verification with Dynamic Probabilistic Assurance, observing an agent's I/O, abstracting it into formal events, learning a Markov Decision Process of the agent's emergent behaviour online, and verifying quantitative PCTL properties in real time, with ...
NIST (2023) [AI RMF 1.0] [notes]	Extraction notes carry substantive analysis; see [notes] for the paper's role in the argument.
Paz (2025) [complexity governance white paper, external evidence, unreviewed] [notes]	Positioning paragraph: Paz (2025), in a policy white paper, argues that risk-based AI governance fails structurally because it assumes linear causality and stable system boundaries, whereas AI operates in complex adaptive socio-technical systems where harm is emergent, delayed, and amplified through feedback loops — and that governance should therefore prioritise dynamic system state over static c...
Pitale et al. (2025) [notes]	Main problem: Traditional safety analysis methods (FMEA, FTA) were designed for systems with clear component boundaries and well-understood failure mechanisms.
Porter et al. (2025) [INSYTE] [notes]	Problem: Existing classification frameworks for AI and autonomous systems — particularly unidimensional Levels of Autonomy (LoA) — are rigid, coarse-grained, sector-specific, prone to misuse, and unable to capture the diverse characteristics of modern AI systems, especially agentic AI systems powered by frontier models.
Reuel et al. (2025) [RAI maturity survey, external evidence] [notes]	Positioning paragraph: Reuel et al. (2025), surveying 1,000 organizations across 20 industries and 19 regions against a two-dimensional responsible-AI maturity model, find that while 9% of organizations reach the highest stage of organizational RAI maturity (policies, governance structures, risk processes), only 0.8% reach it operationally, and none reach it on both dimensions — a systematic gap b...
Robles & Mallinson (2025) [UAIGF, abstract-level only, full text pending] [notes]	Extraction notes carry substantive analysis; see [notes] for the paper's role in the argument.
Sahoo (2026) [notes]	Problem: Military AI governance treats human control as binary (human-in-the-loop or not) and states principles rather than mechanisms.
Saup et al. (2026) [notes]	Main problem: Organisations struggle to move generative AI (GAI) beyond pilot projects into routinised inputs for strategic decision-making (SDM).
Selvam et al. (2026) [notes]	Problem addressed: The literature on AI for process safety in Chemical Process Industries (CPIs) is fragmented across three dimensions: (i)

Paper	Role in the review
	applications are categorised by safety function or algorithm type without linking to autonomy levels or assurance maturity; (ii) "AI for safety" and "safety of AI" are treated as independent subjects; (iii) reviews end with generic recommendations rather than ...
Seong, Lim & Yoon (2025) [CLGuard] [notes]	Extraction notes carry substantive analysis; see [[notes]] for the paper's role in the argument.
Vermaelen & Holvoet (2025) [notes]	Problem addressed: Generating safe and robust behaviour policies for autonomous robots operating in non-deterministic environments — specifically, ensuring that planned actions never lead to unsafe states even when actions produce unintended (alternative) outcomes.
Wu et al. (2025) [single-threaded reasoners, external evidence, preprint] [notes]	Positioning paragraph: Wu et al. (2025), probing three open-weight reasoning LLMs, find that models presented with soft inputs carrying full probability distributions nonetheless rely predominantly on the single highest-probability token at each step, collapsing latent "multi-path" reasoning into a greedy, single-threaded process; exploration is restored only by externally injected randomness (Gum...
Zhao & Yuan (2025) [notes]	Problem addressed: AI healthcare applications have been developed and evaluated primarily in high-income settings.

Formal methods / safety-critical (5)

Formal-methods and safety-critical AI papers that extend or complement the shielding / GS AI machinery in Table A. Included to show that the binary-governance pattern holds across the formal-methods stream, not just the applied one.

Paper	Role in the review
Bach, Kristiansen et al. (2024) [HAI SLR] [notes]	Problem addressed: Human-AI interaction (HAI) in safety-critical industries is essential but poorly understood — the literature is limited, fragmented, and inconsistent.
Klüver et al. (2024) [notes]	Problem: AI methods (especially neural networks and deep learning) are entering safety-critical systems, but existing functional safety standards (EN 61508, EN ISO 13849-1, EN IEC 62061) were not designed for AI.
Mussi et al. (2025) [notes]	Main problem: AI is increasingly integrated into safety-critical network infrastructures (power grids, railway networks, air traffic management), but current AI systems are not designed to optimise the overall efficiency of socio-technical systems or to maximise human performance and engagement.
Newcomb & Ochoa (2026) [notes]	Problem addressed: Traditional testing-based verification is insufficient for providing comprehensive safety guarantees for ML systems deployed in

Paper	Role in the review
	safety-critical domains, due to their non-deterministic, data-driven, black-box nature.
Wang, Poskitt et al. (2025) [Pro2Guard, proactive enforcement, preprint] [notes]	Positioning paragraph: Wang et al. (2025) advance the runtime enforcement line from reactive to proactive: Pro2Guard learns a Discrete-Time Markov Chain of agent behaviour from execution traces and intervenes before violations occur, when the estimated probability of reaching an unsafe state exceeds a user-defined threshold, with configurable intervention modes ranging from halting execution to LL...

Hybrid / neurosymbolic AI (8)

Papers on combining deterministic rule-based systems with probabilistic AI. Included as the theoretical basis for Layer 3 of the proposed architecture (rule-based reasoning engine).

Paper	Role in the review
Belle (2025) [notes]	Extraction notes carry substantive analysis; see [[notes]] for the paper's role in the argument.
Herbold et al. (2024) [contrastive explanations] [notes]	The paper demonstrates that rule-based systems can generate transparent and context-aware explanations, improving user understanding and trust.
Hildebrandt et al. (2026) [XHAILe] [notes]	Extraction notes carry substantive analysis; see [[notes]] for the paper's role in the argument.
Pappula & Rusum (2024) [notes]	The paper demonstrates that hybrid rule-based and machine learning systems improve performance in structured tasks.
Punzi et al. (2024) [notes]	Main problem: The literature on human-AI collaborative decision-making systems is fragmented, with diverse techniques classified sparsely and goals varied across computer science subfields.
Rozenfeld et al. (2026) [GAVEL] [notes]	GAVEL demonstrates that rule-based mechanisms can constrain AI behaviour by monitoring internal activations and applying logical rules.
Sabiri et al. (2025) [hybrid quality recommender SLR] [notes]	Extraction notes carry substantive analysis; see [[notes]] for the paper's role in the argument.
Turgunbaev (2025) [rule-based reasoning] [notes]	Rule-based reasoning provides a deterministic and interpretable foundation for decision-making.

LLM safety and agentic systems (2)

LLM- and agent-specific safety work that complements the primary LLM comparators in Table A (SHIELDAGENT, CRANE, AgentSpec).

Paper	Role in the review
Cash et al. (2025) [LLM confidence judgments, external evidence] [notes]	Positioning paragraph: Cash et al. (2025), in five preregistered studies comparing four LLMs with human participants across aleatory and epistemic uncertainty domains, find that LLM confidence judgments are unpredictably calibrated across domains and models, biased toward overconfidence, and — for ChatGPT and Gemini — insensitive to the models' own past performance: unlike humans, these models fai...
OWASP (2025) [notes]	Problem: Agentic AI systems (agents that plan, decide, and act across multiple steps autonomously) pose novel security risks that go beyond single-model LLM vulnerabilities.

Human-AI interaction and decision-making (2)

Empirical and theoretical studies of how humans and AI share decision authority. Included to inform the human-decision-maker interface and the assumption that the human retains final authority.

Paper	Role in the review
Leppinen et al. (2026) [notes]	Problem: Developing AI solutions for Prognostics and Health Management (PHM) is risky and resource-demanding.
Madsen & Kim (2024) [notes]	Problem: The introduction of maritime autonomous surface ships (MASSs) shifts navigators from direct control to supervising AI collision avoidance systems or managing operations from remote operation centres (ROCs).

Trust and calibration (8)

Trust-measurement and calibration literature. Included to establish that trust-calibration approaches are orthogonal to state-conditioned advisory-scope restriction.

Paper	Role in the review
Aquilino et al. (2025) [notes]	Problem addressed: Trust in AI is measured using highly heterogeneous definitions, instruments, and conceptual frameworks across the literature.
Atf & Lewis (2026) [notes]	Problem addressed: The widely held assumption in XAI research that enhancing explainability in AI systems inherently and substantially increases user trust.
Bach et al. (2024) [HCI trust SLR] [notes]	Problem addressed: User trust in AI-enabled systems is recognised as key to adoption, but there is no consolidated overview of how trust is defined, what factors influence it, and how it is measured — particularly from an HCI (human-centric) perspective focused on the user-AI relationship.
Ghaleb et al. (2026) [notes]	Problem: Vision–Language–Action (VLA) robot policies are overconfident under domain shift; blind trust causes collisions and task failure.
McGrath et al. (2025) [S-TIAS] [notes]	Problem addressed: The Trust in Automation Scale (TIAS; Jian et al., 2000) is the most widely used self-report measure of trust in automated systems, yet it had never been subjected to comprehensive psychometric validation.

Paper	Role in the review
Nastoska et al. (2025) [notes]	Problem addressed: How can trust in AI systems be systematically measured across the AI lifecycle, and what trade-offs arise when optimising for different trustworthiness dimensions (fairness, transparency, privacy, security, accountability)?
Schrills et al. (2025) [notes]	Problem addressed: Self-reported trust questionnaires are the dominant method for assessing trust in human-AI interaction (HAI) research.
Tatasciore & Loft (2025) [notes]	Extraction notes carry substantive analysis; see [[notes]] for the paper's role in the argument.

Low-resource / IoT deployment (6)

Deployment-context papers on low-resource, resource-constrained, and IoT settings. Included to support the deployment-context argument in Chapter 2.

Paper	Role in the review
Bhuvaneswari et al. (2025) [notes]	Problem: Emergency triage in resource-constrained healthcare settings suffers from cognitive biases (anchoring, availability), human error (up to 32% triage error rates), and inability of traditional triage systems (MTS, ESI) to adapt to changing workloads and resource availability.
Di Paco et al. (2026) [AISAFETY] [notes]	This paper demonstrates the effectiveness of rule-based systems for safety monitoring.
Katende (2026) [notes]	Problem: Mainstream AI/ML assumes data abundance and computational scale — assumptions that rarely hold in low-resource environments (LMICs).
Ogenyi et al. (2025) [notes]	Problem: The evolution from traditional IoT to Autonomous IoT (A-IoT) — "self-governing, adaptive, and capable of autonomous decision-making" — creates novel cybersecurity challenges.
Toskov & Toskova (2026) [AgroNova] [notes]	AgroNova demonstrates a practical hybrid architecture where deterministic rule-based control is combined with an AI advisory component.
Zhang et al. (2025) [notes]	Problem: "Many current public safety systems remain encumbered by outdated protocols, manual processes, and fragmented communication channels" (p.1).

Fisheries / maritime domain (22)

Domain evidence for coastal fisheries: empirical studies of fisher decision-making, risk perception, adaptive capacity, and maritime navigation. Included to establish the deployment context and validate the environmental variables in E.

Paper	Role in the review
Atacan & Düzbastılar (2023) [notes]	The paper proactively assesses how small-scale fishermen perceive the risk of navigation under different environmental conditions using a bridge simulator and the Fine-Kinney risk assessment method.
Bossier et al. (2025) [notes]	Problem addressed: While many climate adaptation strategies have been proposed for small-scale fisheries (SSFs), there is a gap between adaptation theory and implementation.
Chandran et al. (2025) [notes]	Problem: Aquaculture faces challenges in efficiency, sustainability, and traceability.
Dominguez-Péry et al. (2023) [notes]	The paper challenges the dominant human-error-centric framing of maritime accident causes.
Gao (2024) [notes]	Main problem addressed: Limited empirical understanding of the nuanced decision-making processes of small-scale fishers, particularly how local context shapes these processes across economic, social, and environmental dimensions.
Haque & Al Jufaili (2026) [notes]	Main problem: AI applications in fisheries and aquaculture are advancing rapidly across species detection, precision aquaculture, IUU fishing detection, and sensing infrastructure, but these advancements are inconsistently deployed, subject to domain shifts, limited by labeled data scarcity, poorly benchmarked, and lack formal uncertainty quantification for decision-relevant outputs.
Jeong & Im (2023) [wave height departure restrictions, Korea small fishing vessels] [notes]	Extraction notes carry substantive analysis; see [[notes]] for the paper's role in the argument.
Kühn et al. (2025) [notes]	Problem addressed: The growing volume and complexity of fisheries data creates a bottleneck requiring automated workflows.
Longobardi et al. (2025) [Peskas] [notes]	Problem addressed: Small-scale fisheries (SSFs) account for over 40% of global fish catch and nearly 90% of fisheries employment, yet suffer from severe data deficiency — particularly in least developed countries — due to logistical, financial, and capacity constraints.
Muhamad et al. (2024) [notes]	Muhamad et al. (2024) validate that environmental, socio-economic, and technological factors are significant determinants of productivity efficiency in Malaysian small-scale fisheries, using a fuzzy model with statistical cross-validation.
Obi et al. (2025) [notes]	Problem addressed: Malaysia's fishery and aquaculture sectors lack a comprehensive, policy-oriented analysis linking production trends with governance, economic constraints, and sustainability challenges.
Rahim et al. (2024) [notes]	Problem addressed: Small-scale fishers in coastal regions face life-threatening decisions about whether to fish during extreme weather

Paper	Role in the review
	events driven by climate change.
Ryu & Han (2025) [notes]	Extraction notes carry substantive analysis; see [[notes]] for the paper's role in the argument.
Shaffril et al. (2017) [notes]	Shaffril et al. (2017) ground the proposed architecture's environmental state vector $E = \{w, r, m, o, v, t\}$ in peer-reviewed evidence from the target domain.
Tahsin et al. (2025) [notes]	Problem addressed: Developing nations face numerous barriers to adopting AI, IoT, and Blockchain in the maritime sector, but existing research focuses on technologically advanced regions and treats barriers in isolation rather than examining their causal interdependencies.
Talpur et al. (2025) [notes]	Problem addressed: Traditional maritime surveillance methods (coastal radar, patrol vessels, satellite imagery, human observation) are insufficient against the volume, stealth, and sophistication of modern maritime security threats including illegal fishing, smuggling, trafficking, and environmental violations.
Tandel et al. (2025) [notes]	Problem: Global aquaculture production exceeds 120 MMT annually (p.2) with annual growth rate exceeding 10%, now accounting for over one-third of global fish consumption (p.1).
Welch et al. (2024) [notes]	Main problem: Roughly 98% of the global fishing fleet (~4.9 million vessels) do not publicly broadcast their locations, creating massive transparency gaps that enable illegal, unreported, and unregulated (IUU) fishing, overfishing, bycatch of protected species, and human rights violations including forced labour.
Wing & Woodward (2024) [notes]	Problem addressed: Fisheries AI innovation and adoption — particularly for electronic monitoring (EM) — faces substantial barriers from fragmented data access, complex regulatory environments, unequal AI capacity across fishing nations, and lack of shared vocabulary between technical and fisheries communities.
Yaakob et al. (2015) [seakeeping stability Malaysian small fishing boats, Tier 1 hydrodynamics] [notes]	Extraction notes carry substantive analysis; see [[notes]] for the paper's role in the argument.
Yamin et al. (2025) [notes]	Yamin et al. (2025) provide the most recent and empirically robust characterisation of the target population for the proposed architecture.
Yuzui & Kaneko (2025) [notes]	Problem: No comprehensive, systematic literature review of MASS risk analysis methods has been previously conducted.

Other / background (7)

Adjacent-domain papers cited for specific background arguments (LLM inference mechanics, complex-systems science, standards). Included where the argument in Chapter 2 draws on non-governance-specific evidence.

Paper	Role in the review
Gabriel et al. (2022) [notes]	Problem: Developing AI-based systems using purely technology-centred approaches fails to capture employee needs, implicit knowledge, and organisational constraints.
Indykov et al. (2025) [notes]	Governance levels implemented: None of the 16 ATs implement Level 1 (participation gate G(S)) or Level 2 (advisory scope restriction A_AI(S)) in the sense used in this architecture.
Kalmykov & Kalmykov (2025) [notes]	Problem: Neural network AI (including XAI) remains fundamentally opaque — a black box whose subsymbolic, statistical nature cannot be made fully transparent.
Li et al. (2026) [notes]	Problem: DRL for autonomous driving motion planning has three safety limitations: (i) sparse collision feedback makes learning safe policies difficult, (ii) safety-critical experiences are diluted in replay buffers, slowing convergence, and (iii) trained policies may still produce reckless actions in unfamiliar scenarios due to extrapolation errors.
Liang et al. (2025) [notes]	Problem: AI models in semantic communication (SemCom) networks for 6G are safety-critical but operate on a "best effort" basis — producing outputs regardless of reliability, exhibiting overconfidence (high confidence scores for incorrect outputs), and lacking generalisation to changing network conditions and knowledge bases.
Wen et al. (2025) [notes]	Problem: As intelligent systems increasingly operate in high-risk environments, understanding how they perceive and respond to hazards is critical — but no systematic study has compared risk perception and human–AI interaction failures across process control systems (PCSs) and autonomous vehicles (AVs).
Yang & Zhu (2024) [industrial expert systems] [notes]	Industrial expert systems demonstrate that rule-based reasoning remains a reliable and interpretable approach for decision support in real-world environments.

Reading rules

- **Ordering.** Both tables are alphabetical by first author, matching Table 2 of [review-protocol.md](#) and Table 3a of [master-coding-table.md](#).
- **Notes link.** Each row's [\[\[notes\]\]](#) link opens the paper's full extraction notes. Follow it for the paper's raw contribution details, methodology, per-dimension governance analysis, and citation trail.
- **Framing convention.** *What the paper does* is written in our framing so that the description maps directly onto our comparison scheme. Papers themselves may use different vocabulary; we prefer to translate consistently.

- **What "Our comment" covers.** Interpretation, not description. What is right about the approach, what is missing relative to our architecture, and — where relevant — why the missing part is load-bearing for our contribution. Comments should be defensible against the paper's own claims; the extraction notes carry the evidence.
- **What "Why we selected it" covers.** The argumentative role the paper plays in the review, in one sentence. Not the paper's importance in general; the paper's importance *to our argument*.